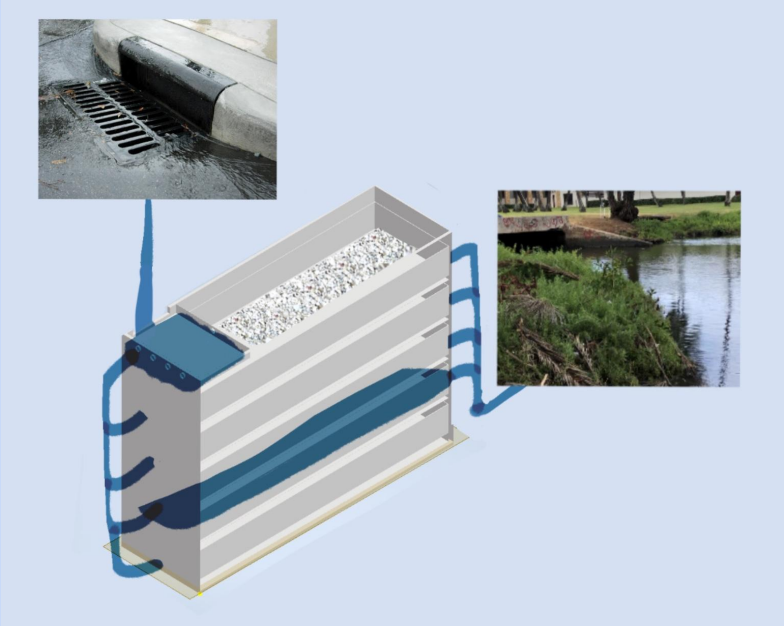
Modular vertical filter:

Urban tamed wetland,
adapted to fit under stormwater grates
and take on exacerbating floods.



Executive summary

[**Need**] Stormwater systems fail to prevent overflow of organic and sediment pollutant into marine ecosystems, damaging:

- Water quality
- Sustainability
- Social and environmental prospects of coastal cities.

[**Solution Features**] Our stormwater filter emulates wetland filtration strategies:

- Redundant layering over vertical-flow column
- De-clogging by biological agents
- Physical multimedia filter

[**Salient benefits**] As compared to competitors, our filter is more:

- Compact thus integratable with existing infrastructure
- Resilient and effective due to larger flow capacity
- Consistent with social and political landscape toward sustainable infrastructure.

The Problem: Background & assumptions

Ala Wai is a living watershed. Urban development has replaced its parts with more:

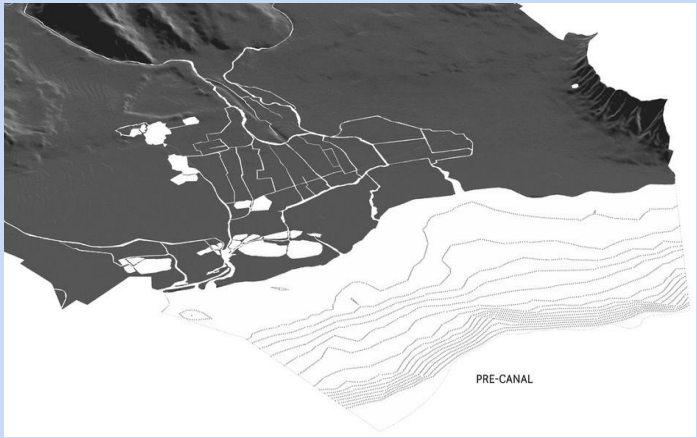
- Tourists
- Residents
- Need for land and water
- Waste

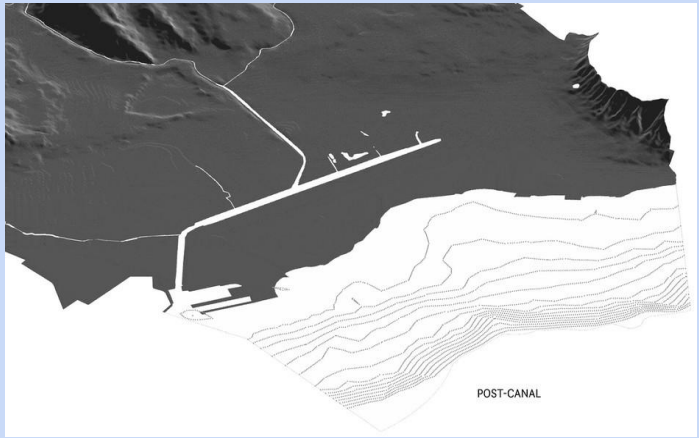
The change is replacing well-flushed wetland within the watershed with:

- An artificial discharge canal
- Tank-pipe treatment transport
- Non-adaptive infrastructure for flood protection.

Fragmented management of these changes means:

- Low resilience to now more violent natural disasters
- Low responsiveness to population and climate trends
- High energy consumption from frequent maintenance
- Toxic outputs





Sean Connelly, 2019

The Problem: Gap analysis

Overall need: The target is Polluted runoff removal for enhancing water quality without compromising flood mitigation.

Stakeholders: city for meeting water quality living standards and reducing maintenance costs, army corps for flood mitigation, environmentalists for pollutant reduction, cultural social activists for reviving traditional sustainability practices against urban development.

Current Solutions: Flood walls and dredging projects are costly and temporary, ignore ecological side effects and use unsustainable materials.

Enabling technology: two converging solutions - large flow capacity physical filters (multimedia gravity rapid sand filters) and biological remediation processes (Modular wetland systems linear)

Problem statement: How might we introduce biological agents into a rapid filter such that we can eliminate pre-filter and bypass components, thus saving space and easing implementation of the more effective technology?

The Problem: design requirements

- Does it remove sufficient pollutants from the stormwater runoff before entering the canal?
- Does the water quality of the coastal watershed improve as a result of the product?
- Does it target municipal wastewater realistically - mostly sediments and nutrient waste?
- Can it handle larger flow volumes and rates than the current system, in scenarios of king tides and 100-year floods? sea level rising?
- Is it more easily monitored and maintained than the current system?
- Is it easily testable and integratable into the current urban infrastructure?

- How might we capture and remove (filter) solid sediments from water?
- How might we capture and remove dissolved organic matter from water?
- How might we distribute water to desired flow rates?
- How might we cycle nutrients captured from water?
- How might we regulate storage of water to prevent overflow? How might we not bypass?

The functional challenge we are solving is filtration vessel space efficiency. It can be represented in fluid dynamics by $P=QR$, $v=Q/A$, where the decision variables are shape of vessel and medium that captures waste, and the target variable is loading rate (pollutant removal rate). Our objectives can be broken down to variable manipulation that we then reference for in biological strategies:

- Reduce flow rate
- Reduce resistance
- Increase surface area
- Increase loading rate

The Solution: concepts considered

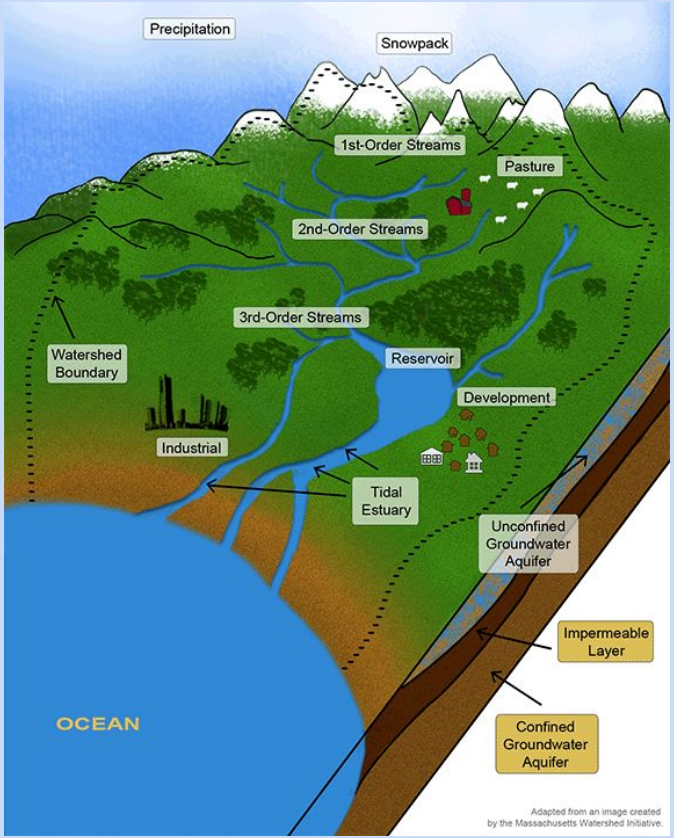
Given the design criteria that we framed into biologized challenges, we explore near and far analogies to the artificial water treatment and transportation system.

Consider watershed system:

- Tributary streams
- Wetlands
- Catch basins
- Groundwater aquifers

Consider analogous filtration Systems at other scales:

- Blood circulation
- Tree symbiosis with microbes, nutrient cycle



The Solution: Conceptual design: concept selection

Patterns that fits with our design requirement criteria:

- Vertical gravity forcing
- Redundant filter modules packed to maximize surface area
- De-clogging by biological processes instead of bypass

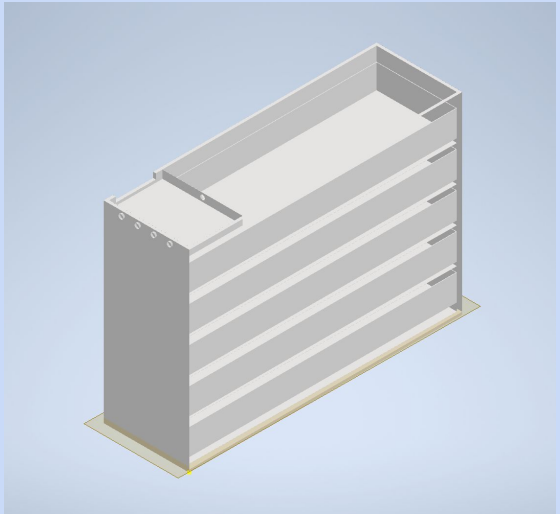
Select the set of strategies associated with vertical-flow wetlands for:

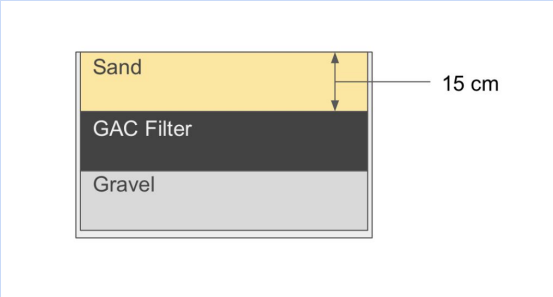
- Ability to deal with extreme flooding events
- Ability to incorporate multiple components of the watershed system into the filter

We study the mechanisms for intercepting, distributing, filtering liquids. for microbe packing and backwashing of filter agents. Literature (Brooks et.al., 2004; Jiang et.al., 2018; Guzman et.al., 2018) highlights key relationship between ordering multimedia in granularity and connecting clusters of media into crossflows aids regular backwash in de-clogging vertical columns.

Key features of our design by these strategies:

- Replacing horizontal-flow wetland filters to a vertical structure





The Solution: Product architecture

To test our prototype we constructed:

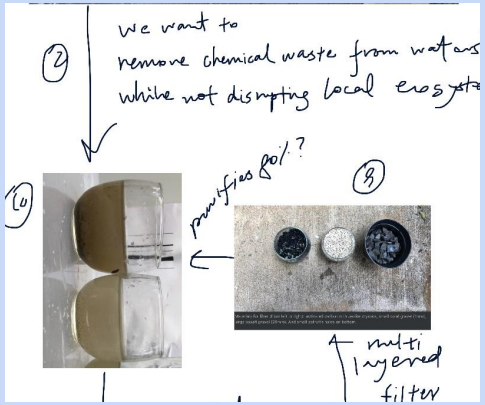
- a **CAD model** with dissemblable and scalable components
- Preliminary **water filtration system**

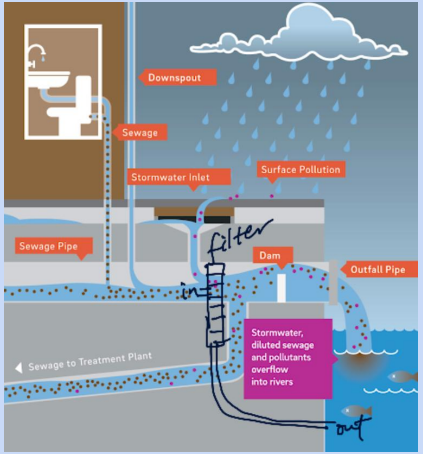
Prototyping process for stormwater system integration:

- Fermi estimation style **calculations**
- Literature **review** on city and EPA budget reports in current flooding and pollution solutions
- **Interviewed** private contractors and investors

Theoretical development from here:

- Systematic experiments for testing
- Hydraulic performance including our design criteria
- Pollutant removal rate
- Flow distribution over surfaces
- Filter material life expectancy under large flows or heavy contamination
- Analysis of neutral effect from biological agents.

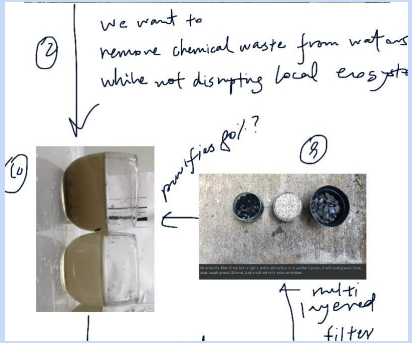




The Solution: Preliminary Evaluation

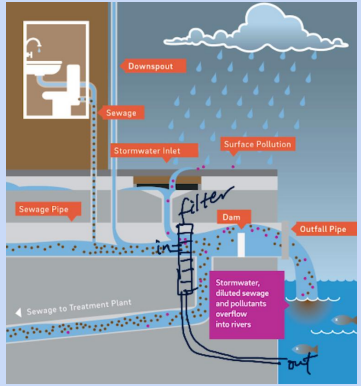
Used prototypes (CAD model and filtration system) to:

- Discuss possible challenges in **interviews** with with working mechanical and chemical engineers
- Conduct preliminary water filtration **experiments** using sampled runoff water in Oahu
- Check approximate quality of purification with the planned materials and smallest scale setup we estimate for easy economic deployment.



The key unknown is integration into stormwater system and below-ground space. Using the Fermi estimation, literature reviews, and interviews we determined:

- Required filter size for extreme flooding events
- Theoretical feasibility of city implementation
- Potential solutions for creating credibility



Stakeholder Interviews

Mechanical and chemical engineers

- Q: how to further distribute flow and pack filter to optimize space use and realistically integrate into stormwater infrastructure? How to get relevant performance testing?
- Our next step: second round of engineering literature review, and MVP product testing to collaborate with labs

Private contractors and investors

- Q: how to practically implement in cities? How to easily service the filter?
- Our next step: Consider best dimensions and materials for ease of shipping, integration with storm drains, and cost effectiveness. Create a cassette system to replace the filter without needing a new structure.

City

- Q: how to get credibility from market for city implementation
- Our next step: input-output analysis, desirability pitches and trials with private companies for initial deployment.

Community and environmental activists

- Q: how to measure effects on pain relieving from better watershed health? How to cooperate with citizen stewards who restore native watershed as cultural practice?
- Our next step: clarify our niche design's role and possible communication, in connection to the solution landscape of citizen initiatives and urban development.

Next Steps:

To make the design feasible we must:

- Create an integration mechanism with current storm drain piping, private company outpipes, etc
- Test various substrates, microbiota, and depths to determine maximum purification levels
- Create a simple maintenance mechanism

This design is worth moving forward with because:

- It has the potential to reduce water pollution through wide scale implementation in cities, farms, and other major polluters
- It is modular, made with cheap materials, and easily replaceable making it a realistic solution

Greatest weak points:

- The concept of using microbiota to clean up after the water has filtered through is not previously tested
- Large initial funding to create a mold or assembly of each part before payout
- GAC filter may not be as effective at removing Flouride, Nitrates, and Sodium

About us

Strengths of this team:

- Small allowing for greater agility in initial stages
- Hard working and curious
- Experience in engineering
- In conversation with school: a global-local perspective that allows complementary analyses and converges with UN goals

Weaknesses to seek help for:

- No past experience bringing a solution to market
  - (Sources: Biomimicry challenge resources, Ellie's connection with a contractor)
- No past experience with the science behind creating microbiomes in filtration systems
  - (Source: Yufei's connection with microbiologist)
- Little experience with product design process
  - (Source: Interest group at mutual college)

Yufei:
Researcher
Analyst
Oahu, HI

Ellie:
Engineer
Researcher
Boston, MA

Two natural science students who want to transform research insights into design solutions.

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